

Estimating Serial Intervals from Outbreak Symptom Onset Data: An Extension of the Vink et al. Transmission-Path Framework

Report Draft V5

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1 Introduction

During the early phase of an outbreak, the most readily observable population-level signal is simply how the number of newly reported cases changes over time. While the speed at which these cases multiply tells us how rapidly the outbreak is expanding, this observed growth rate alone cannot uniquely determine the true transmissibility of the pathogen.

This is because epidemic growth reflects two distinct aspects of transmission. The first is the **magnitude of transmission**, commonly quantified by the reproduction number, R : the average number of secondary infections generated by an infected individual. At the beginning of an outbreak in a fully susceptible population, this quantity is referred to as the basic reproduction number, R_0 . The second is the **timescale of transmission**, described by the generation time: the time between infection in a primary case (the infector) and infection in the corresponding secondary case (the infectee).

These two quantities jointly determine how quickly case numbers grow. For example, the same observed epidemic growth rate could arise from a pathogen that generates relatively many secondary infections over a longer generation time, or from one that generates fewer secondary infections but transmits them more rapidly. Consequently, translating epidemic growth into estimates of the reproduction number requires information about the timing of transmission.

In practice, however, the exact times at which individuals become infected are rarely observed. Generation times are therefore generally not directly measurable. Symptom-onset times are much more readily recorded, making the **serial interval (SI)** a practical observable proxy. The SI is defined as the time between symptom onset in a primary case (the infector) and symptom onset in the corresponding secondary case (the infectee). Although the SI is not identical to the generation time, it is commonly used to inform the transmission-timescale distribution required to estimate reproduction numbers from epidemic growth (Fine 2003; Ganyani et al. 2020). Accurate estimation of the SI is therefore important for quantifying disease transmissibility.

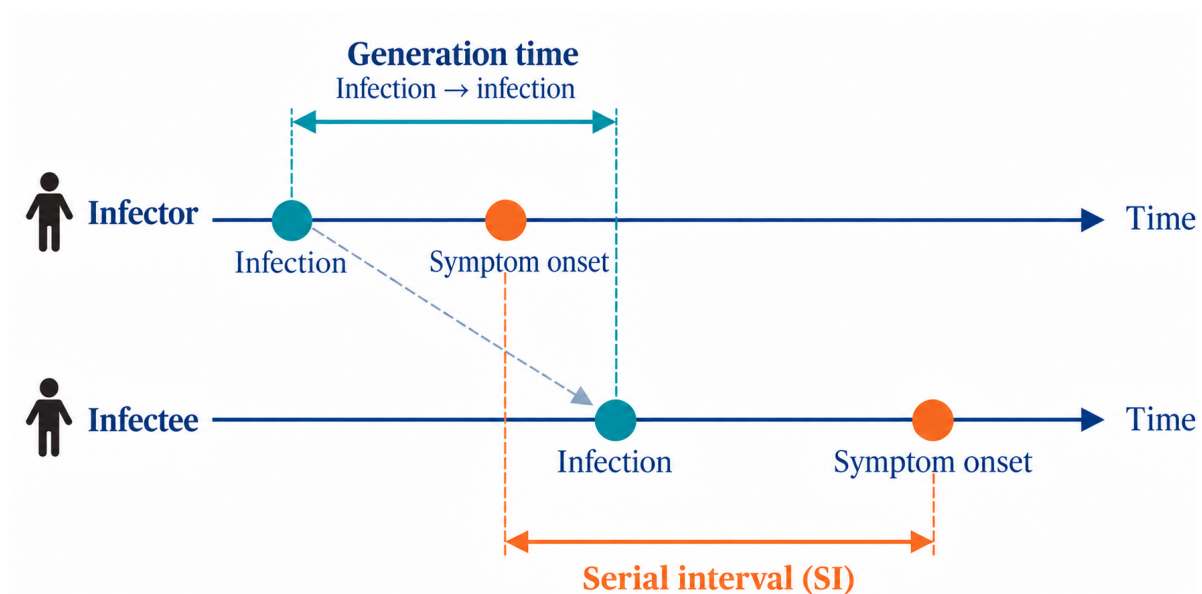


Figure 1: Generation time measures the interval from infection of the infector to infection of the infectee, whereas the serial interval measures the interval between their symptom onsets.

Despite its epidemiological importance, the SI is often difficult to measure directly in real out-break settings. Surveillance data commonly record symptom-onset dates, but the transmission links connecting individual cases are typically unobserved. In other words, researchers may know when each case developed symptoms without knowing exactly who infected whom. As illustrated in panel A of Figure 2, several different transmission networks may therefore be

compatible with the same observed onset data.

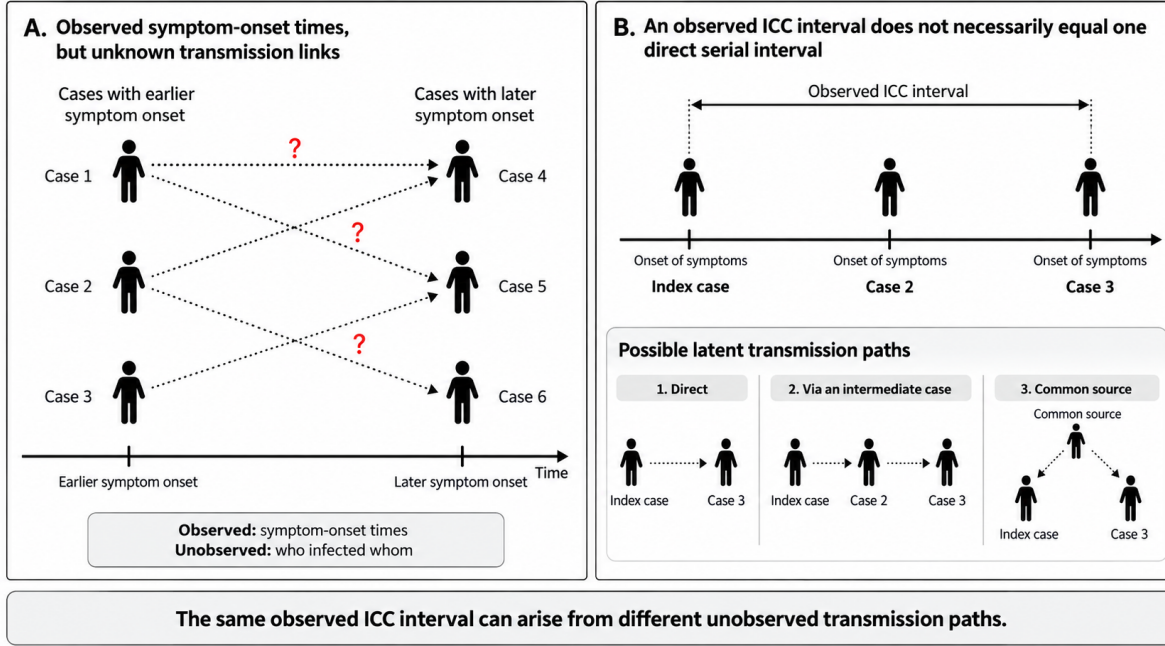


Figure 2: Observed symptom-onset data do not reveal the underlying transmission network. The same index case-to-case (ICC) interval can therefore arise from different latent transmission pathways.

In household or small-cluster settings, the underlying transmission network is typically unobserved. However, one well-defined quantity can still be constructed directly from the available symptom-onset data: the index case-to-case (ICC) interval. By identifying the first recognised symptomatic individual as the index case, the ICC interval for each subsequent case is defined as the elapsed time between that case's symptom onset and the symptom onset of the index case. For a non-index case i , let S_i be that case's symptom-onset time and S_0 the index case's symptom-onset time. The underlying continuous ICC interval is

$$C_i = S_i - S_0.$$

Because the index case is selected as the first symptomatic case, an ICC interval is non-negative. This differs from a direct SI, which can be negative when the infectee develops symptoms before the infector. Importantly, calculating an ICC interval requires only observed symptom-onset times and does not require the underlying infector-infectee pair to be known. It therefore provides a practical way to summarise the timing information available from household or small-cluster outbreak data.

However, an ICC interval is not necessarily a direct serial interval. The ICC is defined relative to the index case, whereas the SI is defined specifically between a true infector and their infectee. Because the underlying transmission chain is unobserved, the same ICC interval may be compatible with several different transmission pathways, including direct transmission, transmission through intermediate cases, or infection from a common source, as illustrated in panel B of Figure 2. Each pathway implies a different relationship between the ICC interval and the underlying SI.

Consequently, symptom-onset data alone cannot determine which transmission pathway generated a particular ICC observation. Estimating the underlying SI distribution requires a statistical framework that accounts for uncertainty in these latent transmission pathways. The following section introduces the mixture-model framework proposed by Vink et al. (2014) to address this problem.

2 Existing Method: Vink et al.

Because the transmission pathway associated with each ICC interval is unknown, an empirical ICC dataset may combine intervals generated through different numbers or configurations of transmission steps. Vink et al. (2014) addressed this ambiguity by modelling the ICC distribution as a finite mixture whose components correspond to four epidemiologically interpretable latent transmission pathways (Vink et al. 2014). This construction links the ICC intervals to the underlying SI distribution while accounting for uncertainty about who infected whom. The following methodology describes how these four pathway-specific components are constructed and combined.

2.1 The Original Normal Mixture Model

In the model of Vink et al., let X denote an underlying serial interval, defined as the time between symptom onset in a primary case and symptom onset in the corresponding secondary case. The serial interval is assumed to follow a Normal distribution,

$$X \sim N(\mu, \sigma^2),$$

where μ is the mean serial interval and σ^2 is the variance of the serial interval.

The Normal distribution provides both epidemiological flexibility and mathematical convenience. Its support over the entire real line allows negative serial intervals, which can occur when an infectee develops symptoms before their infector. In addition, sums and differences of independent Normal variables remain Normal, giving closed-form expressions for the pathway-specific distributions and avoiding the need for numerical integration.

2.1.1 Four latent transmission paths

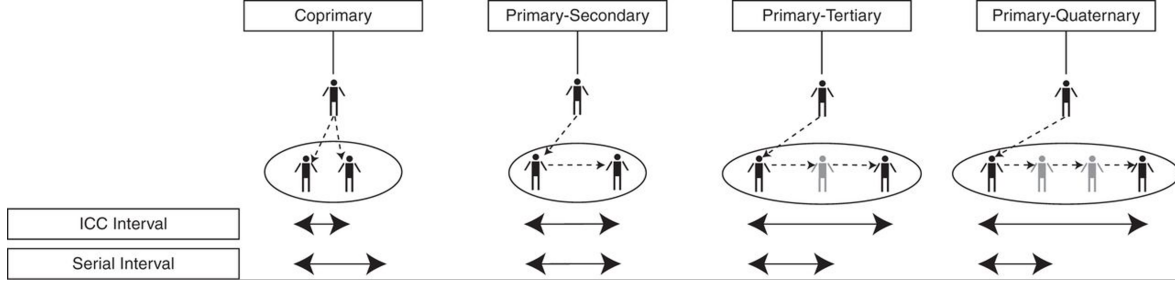


Figure source: Vink et al. (2014)

Figure 3: The four latent transmission pathways linking observed ICC intervals to the underlying serial intervals: co-primary, primary-secondary, primary-tertiary, and primary-quaternary transmission (adapted from Vink et al. (Vink et al. 2014)).

Let

$$\mathcal{G} = \{CP, PS, PT, PQ\}$$

denote the set of four possible latent transmission paths, and let $Z_i \in \mathcal{G}$ denote the unobserved path associated with observation i . Each pathway represents a distinct transmission history and therefore implies a different mathematical relationship between the ICC interval and the underlying serial intervals. Because ICC intervals are non-negative by definition, the path-specific ICC quantities are expressed as absolute values of the relevant sums or differences of Normal serial intervals. These transformations give rise to four component densities, denoted by f_j

- **Co-primary transmission (f_1):** The index case and the other observed case are independently infected by the same person outside the household. The ICC interval is the absolute difference between two independent serial intervals from this common source,

$$C_{CP} = |X_1 - X_2|, \quad X_1 - X_2 \sim N(0, 2\sigma^2).$$

- **Primary-secondary transmission (f_2):** The index case directly infects the other observed case. The ICC interval therefore arises from one direct SI,

$$C_{PS} = |X_1|, \quad X_1 \sim N(\mu, \sigma^2).$$

- **Primary-tertiary transmission (f_3):** The index case infects an intermediate case, which then infects the other observed case. The ICC interval spans two successive serial intervals,

$$C_{PT} = |X_1 + X_2|, \quad X_1 + X_2 \sim N(2\mu, 2\sigma^2).$$

- **Primary-quaternary transmission (f_4):** Transmission from the index case to the other observed case occurs through two intermediate cases. The ICC interval spans three successive serial intervals,

$$C_{PQ} = |X_1 + X_2 + X_3|, \quad X_1 + X_2 + X_3 \sim N(3\mu, 3\sigma^2).$$

Because the transmission path responsible for an individual ICC observation is unobserved, the overall ICC distribution is represented as a finite mixture of these four path-specific components. Let $c \geq 0$ denote a possible value of the continuous ICC interval, and let w_j denote the mixture weight for component f_j . The weights satisfy

$$w_j \geq 0, \quad \sum_{j=1}^4 w_j = 1,$$

the overall continuous ICC density is formulated as the finite mixture

$$f(c \mid \mu, \sigma^2, \mathbf{w}) = w_1 f_1(c \mid \sigma^2) + w_2 f_2(c \mid \mu, \sigma^2) + w_3 f_3(c \mid \mu, \sigma^2) + w_4 f_4(c \mid \mu, \sigma^2).$$

All four mixture components are derived from the same underlying SI distribution and therefore share the common parameters μ and σ^2 . The mixture weights represent the relative contribution of the different transmission pathways to the ICC data.

2.1.2 Discretisation of symptom-onset days

The mixture model above describes the ICC interval on a continuous time scale, whereas the data available in practice are usually recorded only as integer numbers of days. The purpose of this discretisation step is therefore to connect the continuous ICC model to the observed data by deriving the probability of observing an ICC interval of exactly t days.

Let C_i denote the underlying continuous ICC interval for observation i , and let T_i denote its observed integer-day counterpart, with observed value t_i . For example, if the index case is recorded as developing symptoms on 1 May and a later case on 4 May, then the recorded ICC interval is $T_i = 3$ days. However, the exact onset time of either case within 1 May or 4 May is unknown. Thus, both endpoints of the interval are only known to lie somewhere within their respective 24-hour periods. This form of observation is commonly referred to as **doubly interval-censored data** (Vink et al. 2014).

Following Vink et al. (Vink et al. 2014), the exact onset time is assumed to be uniformly distributed within the reported day. Let D_{1i} and D_{2i} denote the recorded onset days of the

index case and the later case. Their exact onset times can be written as $S_{1i} = D_{1i} + U_{1i}$ and $S_{2i} = D_{2i} + U_{2i}$, where $U_{1i}, U_{2i} \stackrel{\text{iid}}{\sim} \text{Uniform}(0, 1)$ represent the unknown within-day onset times.

The recorded integer-day difference is $T_i = D_{2i} - D_{1i}$, whereas the true continuous ICC interval is

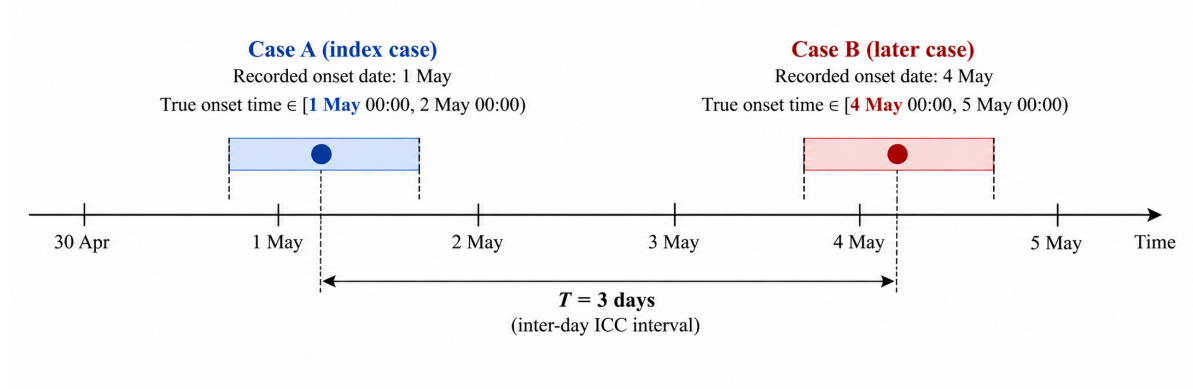


Figure 4: Illustration of doubly interval-censored symptom-onset data. Each recorded onset date identifies a 24-hour interval containing the unknown true onset time, while the observed integer-day ICC interval is the difference between the two recorded dates.

$$C_i = S_{2i} - S_{1i} = T_i + (U_{2i} - U_{1i}).$$

Let $W_i = U_{2i} - U_{1i}$. Because it is the difference between two independent $\text{Uniform}(0, 1)$ variables, W_i follows the symmetric triangular density

$$g(w) = \begin{cases} 1 - |w|, & |w| \leq 1, \\ 0, & |w| > 1. \end{cases}$$

The density is highest at $w = 0$, corresponding to the two cases developing symptoms at approximately the same time of day within their respective recorded days. It decreases linearly towards zero as w approaches -1 or 1 , which would require one onset to occur near the beginning of a day and the other near the end.

Therefore, if the observed interval is $T_i = t_i$, the corresponding continuous interval C_i may lie between $t_i - 1$ and $t_i + 1$. For example, an observed ICC interval of 3 days is compatible with continuous intervals between 2 and 4 days.

To calculate the probability of observing exactly t_i days under the continuous model, the triangular density is shifted to be centred at t_i . Define

$$g_{t_i}(c) = \begin{cases} 1 - |c - t_i|, & |c - t_i| \leq 1, \\ 0, & |c - t_i| > 1, \end{cases}$$

where $g_{t_i}(c)$ gives the weight associated with a possible continuous ICC value c when the recorded interval is t_i days.

Let $f(c \mid \mu, \sigma^2, \mathbf{w})$ denote the continuous ICC mixture density derived in the previous section. The required discrete probability is obtained by integrating this continuous density over all values of c compatible with the observed integer interval, weighted by the triangular kernel:

$$P(T_i = t_i \mid \mu, \sigma^2, \mathbf{w}) = \int_{\max(0, t_i - 1)}^{t_i + 1} f(c \mid \mu, \sigma^2, \mathbf{w}) g_{t_i}(c) dc.$$

The lower limit is truncated at zero because ICC intervals have non-negative support. This equation provides the key link between the continuous model and the discrete data: it converts the continuous ICC density into the probability of observing an integer-day interval of exactly t_i days.

Since the continuous ICC density is itself a mixture,

$$f(c \mid \mu, \sigma^2, \mathbf{w}) = \sum_{j=1}^4 w_j f_j(c \mid \mu, \sigma^2),$$

the same discretisation can be applied separately to each transmission-path component. Define

$$p_{ij}(\mu, \sigma^2) = \int_{\max(0, t_i - 1)}^{t_i + 1} f_j(c \mid \mu, \sigma^2) g_{t_i}(c) dc$$

as the probability of observing t_i days if observation i arose from transmission pathway j . By linearity of integration, the overall probability can then be written as

$$P(T_i = t_i \mid \mu, \sigma^2, \mathbf{w}) = \sum_{j=1}^4 w_j p_{ij}(\mu, \sigma^2).$$

Thus, each observed ICC interval is linked to the continuous model through four path-specific discrete probabilities, which are then combined using the mixture weights.

Once a model-based probability has been obtained for every observed interval, these probabilities can be combined across the full dataset. Let $\mathbf{t} = (t_1, \dots, t_n)$ denote the n observed ICC intervals. Assuming that the observations are independent, their joint probability under

a given set of model parameters is the product of the individual probabilities. When viewed as a function of the unknown parameters with the observed data held fixed, this joint probability defines the likelihood:

$$L(\mu, \sigma^2, \mathbf{w}; \mathbf{t}) = \prod_{i=1}^n \left\{ \sum_{j=1}^4 w_j p_{ij}(\mu, \sigma^2) \right\}.$$

The likelihood therefore measures how well a proposed set of SI parameters and mixture weights explains the observed integer-day ICC data. Parameter estimation seeks values that maximise this likelihood. Because the transmission pathway associated with each observation remains unobserved, direct maximisation is naturally formulated as a latent-variable problem, motivating the expectation–maximisation algorithm described in the next section.

2.2 EM algorithm

Because the true transmission route underlying each ICC observation is not observed, estimation is naturally formulated as a latent-variable problem. Vink et al. addressed this using an expectation–maximisation (EM) algorithm. At each iteration, the E-step computes the posterior pathway responsibilities for every ICC observation, and the M-step updates the mixture weights and SI parameters accordingly. A schematic overview is shown in Figure 5; the detailed updating equations are given in the Appendix.

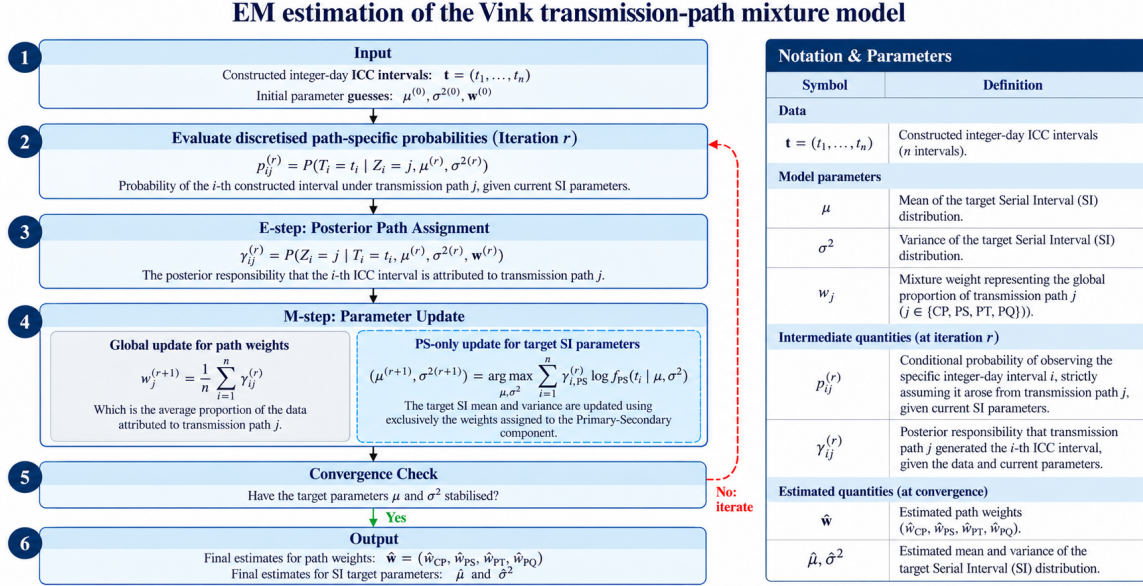


Figure 5: Schematic overview of EM estimation for the original Vink transmission-path mixture model.

2.3 Limitations

While the Vink framework provides a practical approach for estimating SI parameters without known transmission pairs, its performance depends on several modelling assumptions. Two limitations are particularly relevant here: the assumed distributional form of the underlying SI and the restricted number of transmission paths represented in the mixture model.

2.3.1 Distributional assumption for underlying SI

A primary limitation of the Vink framework is its reliance on a prespecified distribution for the underlying serial interval (SI). In the 2014 formulation, the SI is assumed to follow a symmetric Normal distribution, which conveniently allows negative SI values when an infectee develops symptoms before their infector. The non-negative ICC intervals are then constructed from absolute sums or differences of these underlying SI variables and subsequently discretised for symptom-onset dates (Vink et al. 2014).

The Normal assumption is mathematically convenient, but its symmetry may become restrictive when the true SI distribution is asymmetric or has a pronounced upper tail. Serial-interval shape is not universal across infectious diseases. For example, the directly reconstructed serial intervals of primary pneumonic plague were strongly positively skewed, with a reported skewness of 2.3 and a Gamma distribution used to describe their shape (Nishiura et al. 2006). In an early analysis of COVID-19 infector–infectee pairs, a Log-Normal distribution provided the best-fitting serial-interval model among the candidate distributions considered (Nishiura et al. 2020). These examples demonstrate that right-skewed SI distributions can arise in practice, although they do not imply that all serial intervals are right-skewed.

The choice of distribution has also varied within applications of the Vink framework itself. Vink’s earlier thesis assumed a Gamma SI distribution because of its flexible shape (Vink 2010), whereas the 2014 analysis used a Normal distribution as the primary specification and repeated the analysis under a Gamma assumption as a sensitivity analysis, obtaining broadly similar results (Vink et al. 2014). More recently, Ainslie et al. applied the framework to scabies outbreaks using a Normal SI distribution and found that a Gamma sensitivity model provided a poorer fit to those data (Ainslie et al. 2025). Thus, there is no empirical basis for treating a right-skewed distribution as universally preferable to the Normal assumption.

Taken together, these findings suggest that the distributional form of the underlying SI should be regarded as a modelling assumption rather than a fixed feature of the Vink framework. While the Normal specification is convenient and performs adequately in some settings, other distributions may be more appropriate when the SI is asymmetric, strongly right-skewed, or otherwise departs from Normality. The suitability of alternative distributional assumptions therefore remains an open methodological question and warrants further investigation.

2.3.2 Number of transmission paths in the mixture model

A further limitation concerns the restricted set of transmission pathways represented in the mixture. Vink et al. considered four components—co-primary, primary-secondary, primary-tertiary, and primary-quaternary—and excluded longer transmission chains on the grounds that they were unlikely in household settings (Vink et al. 2014). This is a reasonable simplification for small households or similarly contained outbreaks, for which the framework was originally developed. However, the four-path structure may be less adequate in larger clusters or more complex transmission settings, where additional generations of transmission or more complicated transmission histories may occur. The applicability of the model is therefore partly determined by whether these four pathways provide a reasonable approximation to the transmission structure of the outbreak being analysed.

2.4 Motivation for the current project

The Vink framework provides a practical way to infer the underlying serial interval (SI) distribution from constructed ICC intervals by modelling them as a mixture of plausible transmission paths. Existing implementations have primarily relied on Normal and Gamma assumptions for the underlying SI (Vink et al. 2014; Ainslie et al. 2025). These distributions provide convenient and useful starting points, but no single distributional form is expected to describe transmission timing equally well across all infectious diseases. As discussed above, empirical SI distributions can vary substantially in shape, and positively skewed serial intervals have been reported in some settings (Nishiura et al. 2006, 2020). This motivates exploring an additional right-skewed distributional assumption. The Log-Normal distribution was selected as a natural starting point to investigate whether it can provide improved fit or parameter recovery when the underlying SI is positive and right-skewed.

A second motivation is to broaden the practical and methodological flexibility of the existing implementation R package. The `{mitey}` package provides an implementation of the Vink approach, with the current SI modelling framework centred on Normal and Gamma specifications. Extending it to include a Log-Normal option would give researchers a wider choice of distributional assumptions when analysing diseases with different transmission-time characteristics. More broadly, the Log-Normal case provides a useful proof of concept for extending the transmission-path framework beyond distributions for which the path-specific components are analytically convenient. Developing a workable estimation strategy in this setting may therefore provide a foundation for future extensions to other plausible SI distributions, including models that require numerical integration or optimisation.

To address these aims, the following Methods chapter first reconstructs the ideal lognormal mixture framework, then introduces tractable approximations for its non-closed-form components and develops a robust, customized EM procedure to resolve the resulting numerical singularities. The chapter will finally conclude by showing how empirical data and a simulation study were used to test plausibility of the method from parameter-recovery performance.

3 Methods

3.1 Generalisation beyond the Normal distribution

The present extension therefore retains the same four latent transmission paths, mixture weights, ICC construction, and triangular discretisation model introduced in the previous chapter, while changing only the assumed distribution of the underlying SI.

To maintain consistency with the preceding notation, μ and σ^2 continue to denote the mean and variance of the underlying SI on the original time scale. Likewise, the four transmission-path component densities remain denoted by $f_1, \dots, f_4$, with mixture weights $w_1, \dots, w_4$, and p_{ij} continues to denote the discretised probability of the i -th ICC interval under transmission path j .

3.1.1 Ideal Log-Normal mixture framework

Let X_k denote successive direct serial intervals. Under the proposed extension,

$$X_k \stackrel{\text{iid}}{\sim} \text{Lognormal}(\mu, \sigma^2), \quad k = 1, 2, 3, \dots,$$

where μ and σ^2 are defined as the mean and variance of X_k on the original time scale. Equivalently, the corresponding log-scale distribution is

$$\log X_k \sim N \left(\log(\mu) - \frac{1}{2} \log \left(1 + \frac{\sigma^2}{\mu^2} \right), \log \left(1 + \frac{\sigma^2}{\mu^2} \right) \right).$$

Let $f_X(x \mid \mu, \sigma^2)$ denote the resulting Log-Normal density of a direct SI. The four transmission paths retain exactly the same epidemiological interpretation as before, but their path-specific distributions now arise from independent Log-Normal serial intervals:

$$C_{\text{CP}} = |X_1 - X_2|, \quad C_{\text{PS}} = X_1, \quad C_{\text{PT}} = X_1 + X_2, \quad C_{\text{PQ}} = X_1 + X_2 + X_3.$$

Because the Log-Normal distribution has positive support, the PS, PT, and PQ intervals are naturally non-negative and no folding is required for these three components. The absolute value remains necessary for the CP path because its ICC interval is defined by the difference between two independent serial intervals.

3.1.2 Computational difficulty and lack of closed form

The main computational challenge of the Log-Normal extension arises from the loss of the analytical closure available under the Normal assumption. While sums and differences of independent Normal variables remain Normal, the corresponding operations for Log-Normal variables generally do not lead to distributions with convenient closed-form densities. In particular, the CP component involves the absolute difference of two independent Log-Normal variables, whereas the PT and PQ components involve sums of two or three Log-Normal variables. Their exact densities therefore require numerical integration or convolution.

This becomes important for estimation because the path-specific densities must subsequently be discretised through the triangular observation model and evaluated repeatedly throughout the likelihood and EM algorithm. Direct use of the exact Log-Normal expressions would therefore introduce substantially greater computational cost than in the original Normal framework.

The following sections consider each transmission path separately and describe the computational formulation used to obtain tractable path-specific component densities while preserving the underlying transmission-path interpretation.

3.1.3 Path-specific formulation under the Log-Normal extension

The four transmission paths retain the same epidemiological interpretation as in the original Vink framework. However, changing the underlying SI distribution from Normal to Log-Normal alters the mathematical form of several path-specific components. Table Table 1 summarises the ideal construction and the working formulation used for each transmission path.

Table 1: Path-specific formulation under the Log-Normal SI assumption.

Transmission path	Ideal construction	Approximation required?	Working formulation
Co-primary (CP)	$C_{CP} = X_1 - X_2 $	Yes	Half-Normal approximation
Primary-secondary (PS)	$C_{PS} = X_1$	No	Exact Log-Normal
Primary-tertiary (PT)	$C_{PT} = X_1 + X_2$	Yes	Fenton–Wilkinson approximation
Primary-quaternary (PQ)	$C_{PQ} = X_1 + X_2 + X_3$	Yes	Fenton–Wilkinson approximation

The PS component remains available in closed form because it consists of a single Log-Normal SI. In contrast, the CP component involves a difference of Log-Normal variables, while PT and

PQ involve sums of Log-Normal variables; these do not have convenient closed-form densities. Computationally tractable approximations are therefore introduced for these three components.

3.1.3.1 Co-primary transmission: Half-Normal approximation

Under co-primary transmission, the index case and the other case are independently infected by the same external source. The ideal ICC interval is therefore $C_{\text{CP}} = |X_1 - X_2|$, where X_1 and X_2 are independent draws from the same underlying Log-Normal SI distribution.

The absolute difference of two independent Log-Normal variables does not have a convenient closed-form density. Direct evaluation would therefore require repeated numerical integration during likelihood evaluation and EM estimation. To avoid this computational burden, the signed difference $X_1 - X_2$ is approximated by a zero-centred Normal distribution with the same variance. Since

$$\text{Var}(X_1 - X_2) = 2\sigma^2,$$

the working approximation is

$$C_{\text{CP}} \approx |N(0, 2\sigma^2)|,$$

which corresponds to a Half-Normal distribution with scale $\sqrt{2}\sigma$. Its density is

$$f_1(c | \sigma^2) = \frac{1}{\sigma\sqrt{\pi}} \exp\left(-\frac{c^2}{4\sigma^2}\right), \quad c \geq 0.$$

This approximation retains the variance scale of the signed difference before folding and naturally places substantial density near zero, which is consistent with co-primary cases having closely spaced symptom-onset times. However, it is a working approximation rather than the exact distribution of $|X_1 - X_2|$, and its shape depends only on σ^2 , whereas the exact Log-Normal difference distribution depends on both μ and σ^2 .

3.1.3.2 Primary-secondary transmission: Exact Log-Normal component

Under primary-secondary transmission, the index case directly infects the other case, so the ICC interval consists of a single direct serial interval: $C_{\text{PS}} = X_1$.

No approximation is required for this component. Under the assumed Log-Normal SI model,

$$f_2(c | \mu, \sigma^2) = f_X(c | \mu, \sigma^2), \quad c > 0.$$

The PS component therefore remains the exact target SI distribution. As in the original Vink framework, it is this component that provides the direct information used to estimate the SI mean μ and variance σ^2 .

3.1.3.3 Primary-tertiary transmission: Fenton–Wilkinson approximation

Under primary-tertiary transmission, the ICC interval contains two successive direct serial intervals, so $C_{PT} = X_1 + X_2$. Its exact natural-scale mean and variance are 2μ and $2\sigma^2$, respectively.

Although these moments are available directly, the sum of two independent Log-Normal variables does not have a convenient closed-form density. The PT component is therefore approximated using the Fenton–Wilkinson method, which replaces the exact sum with a single Log-Normal distribution having the same first two natural-scale moments:

$$C_{PT} \approx \text{Lognormal}(2\mu, 2\sigma^2).$$

The Fenton–Wilkinson approximation therefore preserves the exact mean and variance of the two-step transmission interval while providing a density that can be evaluated efficiently during likelihood calculation and EM estimation.

3.1.3.4 Primary-quaternary transmission: Fenton–Wilkinson approximation

Under primary-quaternary transmission, the ICC interval spans three successive direct serial intervals: $C_{PQ} = X_1 + X_2 + X_3$. The corresponding exact natural-scale mean and variance are 3μ and $3\sigma^2$.

As with the PT component, the exact sum of three independent Log-Normal variables does not have a convenient closed-form density. The same Fenton–Wilkinson moment-matching strategy is therefore applied:

$$C_{PQ} \approx \text{Lognormal}(3\mu, 3\sigma^2).$$

This approximation again preserves the exact first two moments of the multi-generation interval while avoiding repeated numerical convolution. Together, the Half-Normal approximation for CP, the exact Log-Normal representation for PS, and the Fenton–Wilkinson approximations for PT and PQ provide the four working component densities used in the subsequent mixture likelihood and EM estimation. ### Mixture model and discretised likelihood

After defining the four path-specific working components, the overall mixture retains exactly the same structure as in the original framework:

$$f(c \mid \mu, \sigma^2, \mathbf{w}) = \sum_{j=1}^4 w_j f_j(c \mid \mu, \sigma^2),$$

with

$$w_j \geq 0, \quad \sum_{j=1}^4 w_j = 1.$$

The triangular discretisation model is also unchanged. For the i -th constructed ICC interval $T_i = t_i$, the discrete probability under transmission path j is

$$p_{ij}(\mu, \sigma^2) = \int_{\max(0, t_i-1)}^{t_i+1} f_j(c \mid \mu, \sigma^2) g_{t_i}(c) dc.$$

The corresponding mixture probability is

$$P(T_i = t_i \mid \mu, \sigma^2, \mathbf{w}) = \sum_{j=1}^4 w_j p_{ij}(\mu, \sigma^2),$$

and the likelihood for the complete set of constructed ICC intervals $\mathbf{t} = (t_1, \dots, t_n)$ is

$$L(\mu, \sigma^2, \mathbf{w}; \mathbf{t}) = \prod_{i=1}^n \left\{ \sum_{j=1}^4 w_j p_{ij}(\mu, \sigma^2) \right\}.$$

The transmission-path interpretation, mixture structure, discretisation model, and target parameters are therefore unchanged from the original Vink framework. The principal modification lies in the path-specific component densities: PS remains exact, whereas CP, PT, and PQ use computationally tractable approximations to the corresponding Log-Normal difference or sum distributions.

3.2 EM estimation for the Log-Normal extension

The Log-Normal extension retains the overall EM structure of the original Vink framework: membership of true transmission path is treated as latent, the E-step evaluates the posterior responsibility of each transmission path for every constructed ICC interval, and the M-step updates the pathway weights and the parameters of the target SI distribution. However, several aspects of the estimation procedure require adaptation to accommodate the Log-Normal SI assumption and the working path-specific components introduced above.

Figure 6 provides an overview of the resulting EM algorithm procedure. At each iteration, the discretised probabilities of the four transmission paths are evaluated using the current SI parameters and pathway weights. These probabilities are combined in the E-step to obtain posterior responsibilities. The mixture weights are then updated using information from all four transmission paths, whereas the target SI parameters are updated using the responsibility assigned to the primary-secondary component, following the pathway-specific updating principle of the original Vink algorithm.

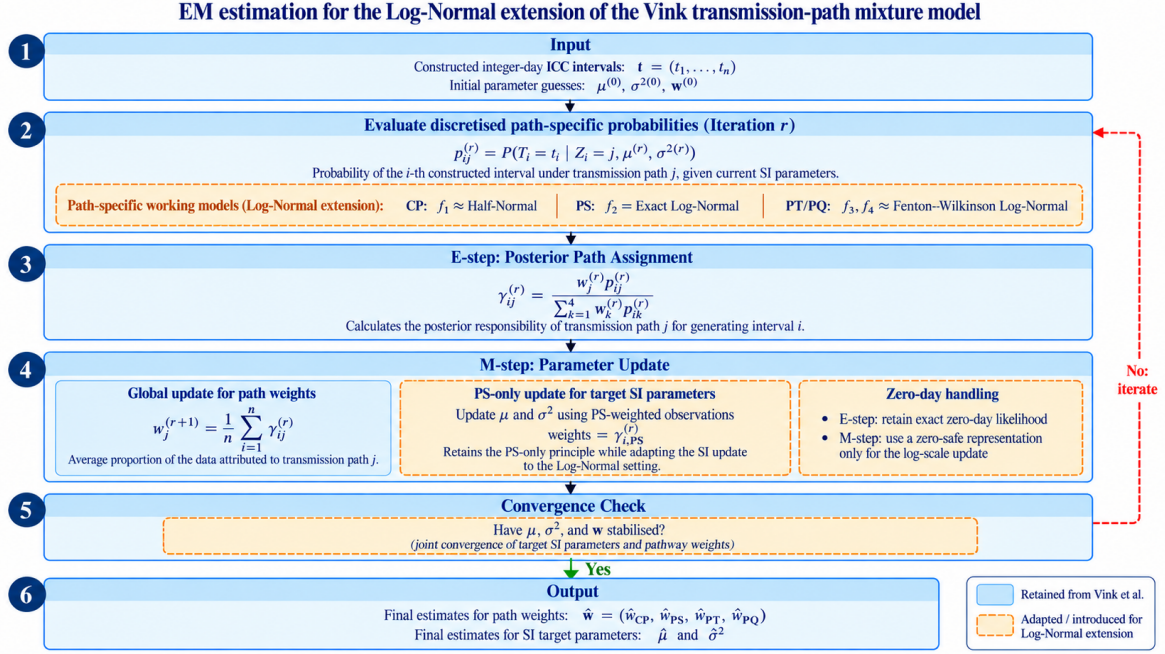


Figure 6: Schematic overview of EM estimation for the Log-Normal extension, showing the retained elements of the Vink procedure and the adaptations introduced for the path-specific working distributions, zero-day observations, and joint convergence assessment.

Initialization and Path-Specific Evaluation. The algorithm initializes with the constructed integer-day ICC intervals alongside initial guesses for the SI parameters and mixture weights. To accommodate the Log-Normal extension, the evaluation of path-specific components is adapted: the primary-secondary (PS) component is evaluated exactly, while the co-primary (CP) component employs a Half-Normal approximation, and the primary-tertiary (PT) and primary-quaternary (PQ) components rely on Fenton–Wilkinson approximations.

E-step: Posterior Path Assignment. The expectation step strictly retains the probabilistic assignment mechanism of the original framework. At each iteration, the current pathway weights are combined with the discretised path-specific probabilities to calculate the posterior responsibility of each latent transmission path (CP, PS, PT, and PQ) for every constructed

ICC interval.

M-step: Parameter Updating and Zero-Day Handling. The maximization step preserves the original structural logic while introducing essential Log-Normal adaptations. First, the global pathway weights are updated by averaging their respective posterior responsibilities across all observations. Second, the algorithm retains the PS-only updating principle—restricting target SI parameter estimation to information weighted by PS responsibilities—but replaces the Normal-scale moment update with a log-scale update, which is then transformed back to the natural-scale mean and variance. To enable this logarithmic transformation without altering path classification, a decoupled zero-day handling strategy is introduced: zero-day ICC intervals retain their exact discretised likelihood in the E-step but are assigned a zero-safe proxy strictly during the logarithmic M-step.

Convergence Assessment. Finally, the convergence criterion is adapted to jointly monitor changes in the target SI parameters and the pathway weights. This dual condition prevents the algorithm from terminating prematurely when the SI estimates appear locally stable but the underlying latent path allocations are still fluctuating.

3.3 Empirical plausibility assessment

The implemented lognormal method was first applied to four scabies datasets previously analyzed under a Normal SI model: studies by Akunzirwe et al. (Akunzirwe et al. 2023), Ariza et al. (Ariza et al. 2013), Kaburi et al. (Kaburi et al. 2019), and Tjon-Kon-Fat et al. (Tjon-Kon-Fat et al. 2021). This analysis is a preliminary numerical plausibility check rather than an independent validation study.

The true SI distributions in these outbreaks are unknown, so the historical data cannot be used to calculate bias, MSE, or parameter-recovery coverage. The limited objective is to determine whether the reconstructed algorithm produces epidemiologically plausible natural-scale means and standard deviations when applied to real, discrete outbreak data. Estimates broadly comparable in magnitude to the earlier Normal-model results would show that the implementation has not failed in an obvious numerical or epidemiological sense. Such agreement would not establish that the lognormal family fits better, nor would it validate its tail shape.

3.4 Simulation study design

The core simulation is the principal controlled assessment of the lognormal procedure. It asks how the method performs with a finite sample of $N = 200$ when both the true natural-scale SI parameters and the pathway composition are known. The design is a 3×3 stress-testing grid rather than an arbitrary collection of settings.

3.4.1 Distribution-shape stress tests

To evaluate model performance under a range of distributional conditions, three natural-scale parameter settings for the SI mean m and standard deviation s were selected:

$$(m, s) \in \{(15, 3), (2, 2), (5, 10)\}.$$

These settings represent progressively more challenging levels of relative dispersion, as measured by the coefficient of variation,

$$c_v = \frac{s}{m}.$$

- **Low dispersion** ($m \gg s$): The setting $(m, s) = (15, 3)$ has a coefficient of variation of 0.2 and serves as the least challenging baseline scenario. Because the mean is substantially larger than the standard deviation, the underlying lognormal SI distribution has relatively low right-skewness. The pathway components are therefore expected to be more clearly separated, providing comparatively favourable conditions for pathway classification and parameter identifiability.
- **Moderate dispersion** ($m \approx s$): The setting $(m, s) = (2, 2)$ has a coefficient of variation of 1. It combines a short mean SI with substantial relative variability, producing greater right-skewness and more probability mass near zero. After the continuously generated ICC values are rounded to integer days, this setting is also expected to produce a larger proportion of zero-day observations. It therefore tests the robustness of the zero-day likelihood treatment, pathway classification, and algorithmic convergence when the observed intervals are small and highly variable.
- **Extreme dispersion** ($s \gg m$): The setting $(m, s) = (5, 10)$ has a coefficient of variation of 2, meaning that the standard deviation is twice the mean. This produces a highly right-skewed SI distribution with a pronounced upper tail and substantial overlap among the pathway components. It serves as an intentionally severe stress test of the Fenton–Wilkinson approximation, latent-path separation, the PS-only parameter update, and the numerical stability of the estimation procedure.

Three natural-scale parameter settings represent progressively different numerical and epidemiological challenges:

$$(m, s) \in \{(2, 2), (15, 3), (5, 10)\}.$$

- $(15, 3)$ is a conventional low-dispersion, low-skewness setting in which component separation should be comparatively clear.

- (2, 2) combines a short mean with substantial relative variability. Rounding places more observations near zero, directly testing the zero-day likelihood and convergence mechanism.
- (5, 10) is an extreme high-dispersion, long-tailed setting. Its coefficient of variation is two, so it deliberately challenges Fenton–Wilkinson tail accuracy, pathway separation, and numerical stability.

3.4.2 Pathway-weight stress tests

Each distributional setting is crossed with three pathway profiles:

Profile	w_{CP}	w_{PS}	w_{PT}	w_{PQ}
Baseline	0.20	0.50	0.20	0.10
CP-heavy	0.40	0.35	0.15	0.10
Higher-generation-heavy	0.10	0.35	0.35	0.20

The baseline profile contains the largest PS share. The CP-heavy profile tests whether the target parameters can be recovered when common-source observations dominate. The higher-generation-heavy profile shifts mass toward PT/PQ, increasing the proportion of observations generated by exact lognormal sums but fitted with moment-matched approximations. Both non-baseline profiles reduce the fraction of direct one-generation evidence and therefore test whether the EM procedure can still isolate m and s from stronger pathway noise.

3.4.3 Data generation and fitting protocol

Within every scenario, the scientific data-generating mechanism uses the unapproximated distributions wherever they define the target model. PS observations are single lognormal draws, while PT and PQ observations are actual sums of two and three independent lognormal draws. CP is generated from the same half-normal component used by the fitted working model. The fitted PT/PQ likelihood uses Fenton–Wilkinson approximations, so the study evaluates both parameter recovery and the practical effect of the sum approximation.

Each of the nine scenarios uses 100 repetitions, seed 2026, a maximum of 50 EM iterations, the default tolerance 10^{-6} , and one starting value initialized at the true (m, s) . The use of an oracle start deliberately removes poor initialization as the primary explanation for failure in this first study; it does not test operational robustness to realistic starts. Pathway counts are fixed as

$$n_g = \text{round}(Nw_g),$$

rather than sampled from a categorical distribution, and the generated continuous ICC intervals are rounded to integer days before fitting.

3.4.4 Evaluation metrics

Convergence is assessed first because an estimate stopped at the 50-iteration cap is not equivalent to a converged estimate. For $\theta \in \{m, s\}$ and $R = 100$ repetitions, the performance summaries are

$$\begin{aligned} \text{Bias}(\hat{\theta}) &= \frac{1}{R} \sum_{r=1}^R \hat{\theta}_r - \theta, \\ \text{MSE}(\hat{\theta}) &= \frac{1}{R} \sum_{r=1}^R (\hat{\theta}_r - \theta)^2, \quad \text{RMSE}(\hat{\theta}) = \sqrt{\text{MSE}(\hat{\theta})}, \end{aligned}$$

and

$$\text{ESD}(\hat{\theta}) = \left\{ \frac{1}{R-1} \sum_{r=1}^R (\hat{\theta}_r - \bar{\hat{\theta}})^2 \right\}^{1/2}.$$

The report also summarizes the convergence rate, terminal iteration, and final pathway weights. For scenarios that do not converge within 50 iterations, bias, MSE, RMSE, and ESD describe iteration-cap terminal states rather than the sampling distribution of a converged estimator. This distinction allows the capped runs to remain diagnostically informative without treating them as successful fits. RMSE and ESD are derived from the saved repetition-level results; no simulation code is rerun for the report.

4 Results

4.1 Simulation results

Convergence within the prespecified 50-iteration limit showed a clear dependence on the distributional setting. The algorithm performed well only in the low-dispersion setting, $(m, s) = (15, 3)$, corresponding to a coefficient of variation of $CV = 0.20$. Under this setting, 99% of repetitions converged for the baseline pathway profile (Scenario 4), 98% for the CP-heavy profile (Scenario 5), and 96% for the higher-generation-heavy profile (Scenario 6). In contrast,

Table 2: Simulation performance for true SI Mean = 2 and SD = 2 (moderate dispersion; N = 200; 100 repetitions)

Scenario	Profile	Parameter	Avg. estimate	Bias	MSE	RMSE	Convergence (%)
1	Baseline	Mean	2.17	0.17	0.13	0.36	0
1	Baseline	SD	1.31	-0.69	0.59	0.77	0
2	CP-heavy	Mean	2.26	0.26	0.16	0.40	0
2	CP-heavy	SD	1.40	-0.60	0.46	0.68	0
3	Higher-generation-heavy	Mean	2.53	0.53	0.45	0.67	0
3	Higher-generation-heavy	SD	1.66	-0.34	0.28	0.53	0

Table 3: Simulation performance for true SI Mean = 15 and SD = 3 (low dispersion; N = 200; 100 repetitions)

Scenario	Profile	Parameter	Avg. estimate	Bias	MSE	RMSE	Convergence (%)
4	Baseline	Mean	15.06	0.06	0.24	0.49	99
4	Baseline	SD	3.10	0.10	0.13	0.36	99
5	CP-heavy	Mean	15.11	0.11	0.37	0.61	98
5	CP-heavy	SD	3.09	0.09	0.15	0.39	98
6	Higher-generation-heavy	Mean	15.14	0.14	0.52	0.72	96
6	Higher-generation-heavy	SD	3.16	0.16	0.27	0.52	96

Table 4: Simulation performance for true SI Mean = 5 and SD = 10 (extreme dispersion; N = 200; 100 repetitions)

Scenario	Profile	Parameter	Avg. estimate	Bias	MSE	RMSE	Convergence (%)
7	Baseline	Mean	3.37	-1.63	3.55	1.88	0
7	Baseline	SD	3.66	-6.34	42.79	6.54	0
8	CP-heavy	Mean	2.98	-2.02	4.63	2.15	0
8	CP-heavy	SD	3.06	-6.94	49.52	7.04	0
9	Higher-generation-heavy	Mean	4.33	-0.67	2.09	1.45	0
9	Higher-generation-heavy	SD	4.99	-5.01	29.36	5.42	0

none of the repetitions converged within 50 iterations under either the moderate-dispersion setting, $(m, s) = (2, 2)$, or the extreme-dispersion setting, $(m, s) = (5, 10)$, irrespective of the pathway profile. The consistency of this pattern across the three pathway compositions suggests that convergence difficulty was driven primarily by the shape and relative dispersion of the underlying SI distribution rather than by pathway composition alone.

For the low-dispersion scenarios that achieved high convergence rates, recovery of the true SI parameters was generally accurate. Under the baseline profile, the estimated SI mean was 15.06 compared with the true value of 15, corresponding to a bias of 0.06 and an RMSE of 0.49. The estimated SI standard deviation was 3.10 compared with the true value of 3, with a bias of 0.10 and an RMSE of 0.36.

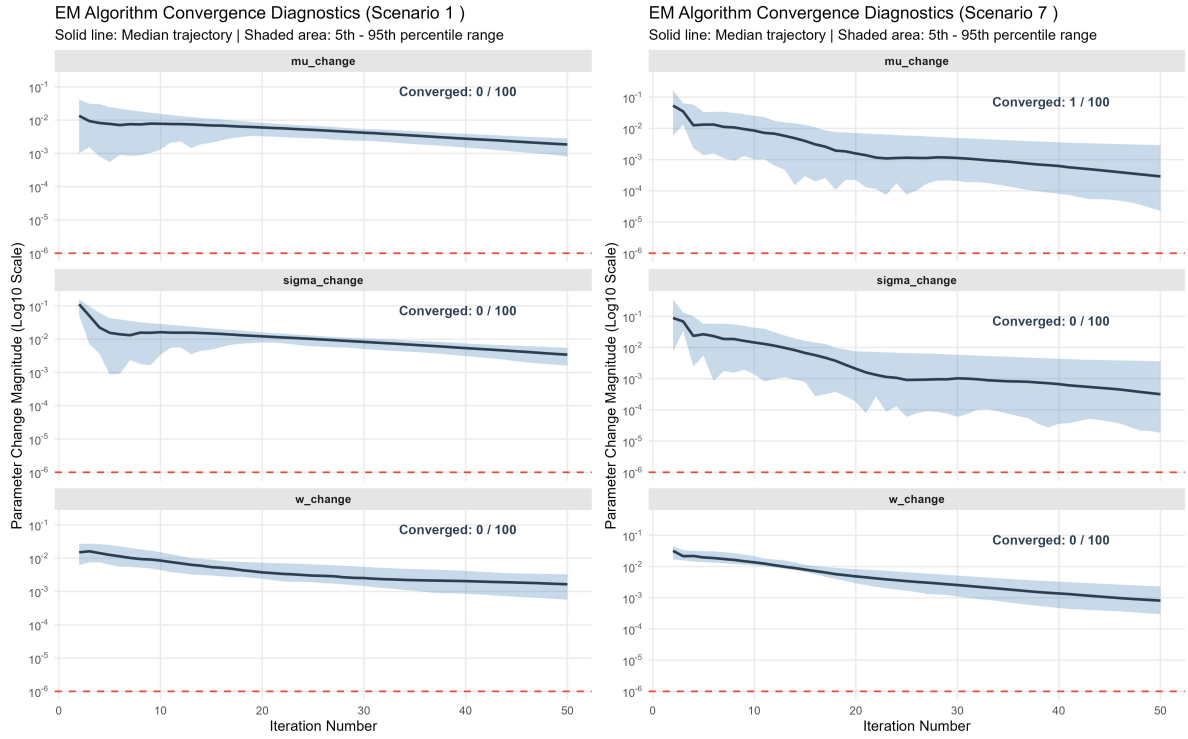
Estimation also remained close to the true values under the CP-heavy and higher-generation-heavy pathway profiles, although precision was lower than under the baseline profile. For the CP-heavy profile, the RMSEs were 0.61 for the SI mean and 0.39 for the SI SD, whereas under the higher-generation-heavy profile they increased to 0.72 and 0.52, respectively. Both profiles contained the same PS proportion of 0.35, yet the higher-generation-heavy profile produced larger estimation errors. This suggests that the reduction in precision cannot be explained by the lower proportion of direct PS observations alone. A greater contribution from PT and PQ observations may introduce additional estimation difficulty, potentially because these pathways provide less direct information about the underlying SI parameters and are represented using working approximations.

The interpretation of Scenarios 1–3 and 7–9 is different because none satisfied the joint convergence criterion within 50 iterations. Their reported estimates therefore represent iteration-capped terminal states rather than results from a converged estimator. At iteration 50, the moderate-dispersion scenarios showed mild overestimation of the SI mean and underestimation of the SI SD, whereas the extreme-dispersion scenarios showed more pronounced underestimation, particularly for the SD. These terminal values are primarily diagnostic and should not be interpreted as evidence of estimator bias until convergence behaviour has been investigated under a larger iteration limit.

To better understand why the non-convergent scenarios had not met the stopping rule by iteration 50, the changes in the SI mean, SI SD, and pathway weights were examined across iterations. The plots show that these quantities generally continued to decrease over time, but at different rates across scenarios.

For Scenarios 1–3, the SI SD changed more slowly than the mean and pathway weights, and remained the furthest from the convergence threshold by iteration 50. In Scenarios 7–8, the pathway weights were slower to stabilize, while in Scenario 9 the mean, SD, and weights remained at broadly similar levels. This suggests that there was no single parameter responsible for non-convergence across all scenarios; instead, the slowest component depended on the distributional setting.

Importantly, the trajectories were still moving downward at iteration 50. Therefore, these results do not yet show whether the algorithm would eventually converge if given more itera-



(a) Scenario 1: moderate dispersion, $(m, s) = (2, 2)$. (b) Scenario 7: extreme dispersion, $(m, s) = (5, 10)$.

Figure 7: Representative convergence diagnostics for non-convergent scenarios under the base-line pathway profile.

tions. The next step is to increase the iteration limit and examine how the convergence rate changes as more iterations are allowed.

Figure 7 shows representative convergence diagnostics for two non-convergent scenarios under the baseline pathway profile. Scenario 1 represents the moderate-dispersion setting, $(m, s) = (2, 2)$, and Scenario 7 represents the extreme-dispersion setting, $(m, s) = (5, 10)$. The solid line shows the median parameter-change trajectory across repetitions, the shaded region shows the 5th–95th percentile range, and the dashed horizontal line denotes the convergence tolerance of 10^{-6} .

Several mechanisms may contribute to this behaviour. As relative dispersion increases, stronger right-skewness and greater overlap between pathway-specific ICC distributions can weaken pathway separation and slow the joint updating of SI parameters and mixture weights. In the extreme-dispersion setting, the larger discrepancy between the exact PT/PQ sum distributions and their Fenton–Wilkinson working approximations may also contribute to poorer stabilization. However, the current 50-iteration results alone cannot distinguish slow convergence from persistent algorithmic instability.

5 Software Implementation

6 Discussion

This study aimed to examine whether the Vink et al. transmission-path mixture framework can be extended by replacing the original Normal or Gamma assumption for the underlying serial interval with an iid lognormal assumption and whether a lognormal extension provide a better statistical fit and more plausible estimates. The current work provides a theoretical answer to the first part of this research question. The four-path ICC mixture structure can be retained under the lognormal extension, and the primary-secondary component remains directly interpretable as the target serial interval distribution.

The main value of this extension is that it try to provide a more biologically plausible option when the serial interval is expected to be strictly positive and right-skewed. In particular, the lognormal distribution avoids negative serial intervals and can accommodate a longer right tail than more Normal and Gamma distributional assumptions, which corresponds to infectious disease settings where delayed transmission or high variability in symptom-onset timing may produce skewed ICC interval data.

However, the results also show that this gain in biological plausibility comes with an important computational cost. Under the Normal assumption, several path-specific distributions are analytically convenient because sums of independent Normal random variables remain Normal. Under the lognormal assumption, this property is lost. The co-primary, primary-tertiary, and primary-quaternary paths involve absolute differences or sums of independent lognormal variables, which do not have simple closed-form density functions. As a result, the likelihood

and EM algorithm require numerical integration and numerical optimisation. Thus, although the lognormal extension is theoretically well defined, its practical implementation is more computationally demanding than the original framework.

6.1 Limitations

The present study provides a proof of concept for extending the Vink transmission-path framework to a lognormal serial interval. However, its theoretical and computational findings should be interpreted in light of several approximation errors, algorithmic compromises, and structural assumptions.

6.1.1 Approximation Costs in Latent Pathways

To maintain computational feasibility, the fitted model replaces several exact pathway distributions with tractable approximations. For the co-primary (CP) pathway, the exact distribution of the absolute difference between two serial intervals depends on both the natural-scale mean (m) and standard deviation (s), whereas the half-normal approximation depends only on s . As a result, the approximation discards information related to m and may distort the curvature and tail behaviour of the CP component, particularly when the underlying SI distribution is highly dispersed.

For the PT and PQ pathways, the Fenton–Wilkinson (FW) approximation matches only the first two moments of the exact sums of lognormal serial intervals. As the coefficient of variation increases, the approximation increasingly understates the right-skewness of these higher-generation distributions. These discrepancies may also affect the E-step indirectly, because pathway-specific working densities determine the posterior responsibilities used to classify observations across the four latent pathways.

6.1.2 Algorithmic Compromises in Parameter Updating

The estimation procedure uses a PS-only parameter update in the M-step. Restricting the update to primary-secondary responsibilities reduces the direct influence of higher-generation observations and their associated approximation errors. The trade-off is that information about the underlying SI parameters contained in the CP, PT, and PQ pathways is not used directly in the parameter update. When the PS proportion is small or when pathway distributions overlap substantially, this restriction may reduce the effective information available for estimating the SI mean and standard deviation.

A fuller parameter update based on multiple or all pathways could, in principle, make use of additional information. However, such an approach would require repeated numerical evaluation

and optimization of more complicated pathway likelihood contributions, substantially increasing computational cost. Under typical settings, the corresponding improvement in estimation accuracy may therefore be limited relative to the additional computational burden.

6.1.3 Model Identifiability and Convergence

The performance of the iterative estimation procedure depends strongly on the degree of separation between pathway components. When the underlying SI distribution has high relative dispersion, the four pathway distributions can overlap substantially. This makes pathway allocation more difficult, weakens practical identifiability, and may lead to slow stabilization of both the SI parameters and pathway weights.

The simulation study also used an oracle starting value, with the initial SI mean and standard deviation set to their true values. This design was useful for isolating the behaviour of the proposed method under favourable initialization, but it does not assess sensitivity to realistic starting values. In practical applications, random or poorly chosen initial values may increase the risk of slow convergence, convergence to different stationary points, or failure to satisfy the stopping criterion.

6.1.4 Simulation Scope and Empirical Verification

The current simulation study is limited in scope. All scenarios used a fixed sample size of $N = 200$, and pathway counts were determined from prespecified proportions rather than sampled randomly. The study therefore does not yet assess how performance changes across different sample sizes or under additional variability in pathway composition.

Empirical assessment is also limited by the absence of known transmission pairs or a known true SI distribution in the historical datasets. Although the proposed model can be fitted to these datasets and produces epidemiologically plausible estimates, the empirical data cannot establish whether the lognormal specification is superior to the Normal or Gamma alternatives. At present, empirical evaluation is therefore restricted to plausibility checks and visual comparison between fitted distributions and the ICC data.

6.1.5 Structural and Observational Assumptions

Finally, the framework retains several structural assumptions inherited from the original transmission-path model. The four-pathway structure is most appropriate for small household or cluster settings and may become restrictive when longer transmission chains, multiple external introductions, or more complex transmission networks are present.

The model also treats ICC observations as statistically independent, which does not account for possible dependence among cases from the same household or outbreak cluster. In addition,

the discretisation model assumes that exact symptom-onset times are uniformly distributed within each reported calendar day. This simplification does not explicitly account for other forms of measurement error, including recall error, reporting delay, or inaccuracies in recorded onset dates.

6.1.5.1 ***** remains to be updated: limitation for co-primary path

This approximation preserves the variance scale of the unfurled difference and provides a normalized density that is inexpensive to evaluate. Its concentration at zero is also consistent with the expectation that cases infected by a common source may develop symptoms within a short time of one another.

The approximation does not reproduce the complete distribution of the absolute difference between two lognormal variables. In the exact construction, the shape of the CP distribution depends on both the natural-scale mean m and standard deviation s of the underlying SI distribution. By contrast, the half-normal approximation depends only on s . It may therefore differ from the exact CP distribution in its curvature and tail behaviour, particularly when the underlying serial intervals are highly dispersed or strongly right-skewed. The approximation should consequently be interpreted as a computationally tractable CP working model rather than an exact distributional identity.

Zero-day observations create a related but distinct problem in the logarithmic M-step because $\log(0)$ is undefined. The half-normal CP approximation provides stable density near zero, but it does not by itself resolve that log-scale singularity. Section 3.3.3 describes the E/M-decoupled strategy used to retain the correct zero-day interval likelihood in the E-step while applying a neutral log-scale representation in the M-step.

6.1.6 ***** remains to be updated : Distributional cost of the F-W approximation.

By construction, the FW approximation reproduces the exact mean and variance of each sum. Its approximation error therefore appears in higher-order characteristics of the distribution, particularly skewness and tail shape.

Let

$$c_v = \frac{s}{m}$$

denote the coefficient of variation of the underlying SI distribution. For a sum of n independent and identically distributed lognormal variables, the skewness of the exact sum is

$$\gamma_{\text{sum},n} = \frac{(c_v^2 + 3)c_v}{\sqrt{n}}.$$

The skewness of the moment-matched FW lognormal distribution is

$$\gamma_{\text{FW},n} = \frac{(c_v^2/n + 3)c_v}{\sqrt{n}}.$$

The resulting skewness gap is

$$\Delta\gamma_n = \gamma_{\text{sum},n} - \gamma_{\text{FW},n} = \frac{c_v^3(1 - 1/n)}{\sqrt{n}}, \quad n > 1.$$

Because $\Delta\gamma_n > 0$, the FW distribution has lower skewness than the corresponding exact sum. The approximation therefore compresses the overall right-skewness of the PT and PQ distributions. This statement concerns the distribution’s standardized third moment; it does not imply that the FW density is lower than the exact density at every point in the right tail.

The magnitude of the discrepancy depends on relative dispersion, $c_v = s/m$, rather than on the absolute time scale alone. The discrepancy increases rapidly with c_v because it is proportional to c_v^3 . Consequently, the approximation is expected to be accurate when the underlying SI distribution has low relative dispersion, but potentially less accurate when the SI is highly variable and strongly right-skewed.

6.2 Future work

The next step is to implement the proposed lognormal extension within the existing `{mitye}` R package framework, which means to replace the original Normal or Gamma SI distribution option with the lognormal specification derived in this study, and then test whether the full SI inference workflow remains computationally feasible.

This implementation will allow us to assess whether the anticipated computational challenges arise in practice. In particular, the CP, PT, and PQ path-specific probabilities may require repeated numerical integration, and the M-step may require numerical optimisation rather than closed-form parameter updates. This part will focus not only on coding the lognormal extension, but also on exploring more efficient numerical methods such as integration approximation if computational difficulties are observed.

A simulation study will then be conducted to evaluate the performance of the proposed method. Under correct specification, data can be generated from an iid lognormal SI distribution, converted into ICC intervals through the four latent transmission paths, and then fitted using the proposed EM algorithm. This would allow assessment of bias, root mean squared error, convergence rate, and the recovery of the SI mean and standard deviation. Under misspecification scenarios, the data-generating SI distribution could be changed to Gamma, Weibull, or another positive distribution to evaluate the robustness of the lognormal extension.

Future work should also compare the lognormal extension with the existing Normal and Gamma versions of the Vink framework given real outbreak datasets of different infectious disease. Such comparisons would help determine when the lognormal assumption provides improved estimation or better model fit. This is important because the lognormal model is more computationally expensive, so its practical value needs to be justified by improved performance in relevant scenarios.

Finally, the transmission-path framework itself could be extended. Future models could include additional transmission generations beyond the current four paths, or incorporate more flexible assumptions about the transmission network. More general SI distributions, such as Weibull, log-logistic, or generalized gamma distributions, could also be explored. These extensions may broaden the applicability of the method to more complex real-world transmission settings.

7 Appendix

7.1 EM estimation

Rationale for the EM algorithm. Although the mixture model defines the possible transmission pathways, the pathway responsible for any individual ICC observation remains unobserved. This creates a latent-variable estimation problem. Assigning every observation to one pathway before fitting the model would ignore this uncertainty and could produce unstable or biased estimates when several pathways are plausible.

The expectation-maximization (EM) algorithm addresses this problem through probabilistic rather than deterministic classification. In the expectation step (E-step), the algorithm calculates the posterior responsibility of each pathway—CP, PS, PT, or PQ—for every observed ICC interval. Each observation can therefore contribute fractionally to several pathways according to its compatibility with their current weights and component distributions. In the maximization step (M-step), these responsibilities are used to update the pathway proportions and the underlying SI parameters.

A distinctive feature of the Vink procedure is that the SI mean and variance are updated using the responsibilities assigned to the PS component only. Because the PS pathway represents direct, one-generation transmission, it contains the least contaminated information about the target SI distribution. Very short intervals that are more compatible with common-source exposure and long intervals that are more compatible with multi-generation transmission tend to receive lower PS responsibilities. They therefore contribute less to the SI parameter updates. This PS-weighted mechanism acts as a pathway-based filter and provides robustness against contamination by non-direct transmission.

Because the path label Z_i is unobserved, Vink et al. estimated the model using an expectation-maximization (EM) algorithm. Introduce the latent indicator

$$z_{ij} = \begin{cases} 1, & Z_i = j, \\ 0, & Z_i \neq j. \end{cases}$$

If the path labels were known, the component-membership contribution to the complete-data log-likelihood would be

$$\ell_c = \sum_{i=1}^n \sum_{j=1}^4 z_{ij} \{ \log w_j + \log p_{ij}(\mu, \sigma^2) \}.$$

E-step: posterior path responsibilities. At iteration r , suppose the current parameter estimates are

$$\mu^{(r)}, \quad (\sigma^2)^{(r)}, \quad \mathbf{w}^{(r)} = (w_1^{(r)}, w_2^{(r)}, w_3^{(r)}, w_4^{(r)}).$$

For each observation i and each candidate path j , the first calculation is the path-specific observed probability

$$p_{ij}^{(r)} = p_{ij}(\mu^{(r)}, (\sigma^2)^{(r)}) = P(T_i = t_i \mid Z_i = j, \mu^{(r)}, (\sigma^2)^{(r)}).$$

This quantity answers a conditional question: *if observation i had arisen from path j , how probable would its observed integer interval t_i be under the current SI parameters?*

The E-step then reverses the conditioning. Its posterior responsibility is defined as

$$\gamma_{ij}^{(r)} = P(Z_i = j \mid T_i = t_i, \mu^{(r)}, (\sigma^2)^{(r)}, \mathbf{w}^{(r)}).$$

Applying Bayes' rule gives

$$\gamma_{ij}^{(r)} = \frac{w_j^{(r)} p_{ij}^{(r)}}{\sum_{k=1}^4 w_k^{(r)} p_{ik}^{(r)}}.$$

The terms have a direct interpretation:

- $w_j^{(r)}$ is the current prior probability, or prevalence, of path j ;
- $p_{ij}^{(r)}$ measures how well path j explains the observed interval t_i ; and
- the denominator is the total probability of observing t_i across all four possible paths.

The numerator therefore combines pathway prevalence with observation compatibility. The denominator normalizes these four competing explanations so that

$$\sum_{j=1}^4 \gamma_{ij}^{(r)} = 1 \quad \text{for every observation } i.$$

Thus, $\gamma_{ij}^{(r)}$ is the fractional membership assigned to observation i for path j , rather than a hard pathway classification. Averaging these responsibilities gives the updated mixture proportions:

$$w_j^{(r+1)} = \frac{1}{n} \sum_{i=1}^n \gamma_{ij}^{(r)}.$$

The parameter update described by Vink et al. then uses the posterior weights for the second, primary-secondary component only:

$$\mu^{(r+1)} = \frac{\sum_{i=1}^n \gamma_{i2}^{(r)} t_i}{\sum_{i=1}^n \gamma_{i2}^{(r)}},$$

and

$$\{\sigma^2\}^{(r+1)} = \frac{\sum_{i=1}^n \gamma_{i2}^{(r)} (t_i - \mu^{(r+1)})^2}{\sum_{i=1}^n \gamma_{i2}^{(r)}}.$$

This PS-only local update acts as a pathway-based outlier filter. Very short intervals that resemble common exposure and very long intervals that resemble multi-generation transmission receive less PS responsibility and are correspondingly downweighted in the SI mean and variance. The mechanism is the source of the procedure's robustness to pathway contamination: it uses all observations to learn pathway membership but allows the most plausible one-generation information to dominate the target-parameter update.

The E-step and parameter updates are repeated until the estimates converge. The article does not state a single numerical tolerance; it reports using several starting values to reduce the risk that the algorithm stops at a local rather than global maximum. Vink et al. also checked the method on simulated data.

7.2 Updated EM algorithm for log-normal extension

7.3 Robust EM algorithm implementation

The fitted procedure retains the EM logic of probabilistic pathway assignment followed by parameter updating, but it is customized for a lognormal target, approximate higher-generation components, and zero-day observations. The modifications are presented as parts of one coherent estimation strategy.

7.3.1 Posterior responsibilities with current pathway weights

For a finite mixture, the posterior probability of pathway g must combine its current prevalence with its compatibility with the observation. At iteration r , let $p_g(T_i; m^{(r)}, s^{(r)})$ be the discretised working likelihood for observation i . The E-step is

$$\tau_{gi}^{(r)} = \frac{w_g^{(r)} p_g(T_i; m^{(r)}, s^{(r)})}{\sum_h w_h^{(r)} p_h(T_i; m^{(r)}, s^{(r)})}.$$

Here, $w_g^{(r)}$ is the prior pathway probability within the current iteration, $p_g(T_i)$ is the pathway-specific likelihood, and $\tau_{gi}^{(r)}$ is the posterior responsibility. Omitting $w_g^{(r)}$ would implicitly reset the four pathways to equal prevalence at every E-step and could create systematic misclassification when their true frequencies differ.

The pathway proportions are updated by

$$w_g^{(r+1)} = \frac{1}{N} \sum_{i=1}^N \tau_{gi}^{(r)},$$

and these updated proportions are returned to the next E-step. This feedback ensures that estimated pathway prevalence and individual classification remain probabilistically coherent.

7.3.2 PS-only weighted-moment update

After the E-step, all pathway proportions are updated using their corresponding posterior responsibilities. However, the parameters of the underlying lognormal SI distribution are updated using only the responsibilities assigned to the primary-secondary pathway. This distinction is important: the procedure updates all mixture weights, but it does not use all four pathways equally when estimating the target SI mean and standard deviation.

Let

$$a_i^{(r)} = \tau_{\text{PS},i}^{(r)}$$

denote the PS responsibility for observation i at iteration r , and let

$$Z_i = \log(T_i^*)$$

denote the corresponding log-scale M-step value. Define

$$A_1^{(r)} = \sum_{i=1}^N a_i^{(r)}, \quad A_2^{(r)} = \sum_{i=1}^N \left(a_i^{(r)}\right)^2.$$

The log-scale SI parameters are updated as

$$\eta^{(r+1)} = \frac{\sum_{i=1}^N a_i^{(r)} Z_i}{A_1^{(r)}}$$

and

$$(\zeta^2)^{(r+1)} = \frac{A_1^{(r)}}{\left(A_1^{(r)}\right)^2 - A_2^{(r)}} \sum_{i=1}^N a_i^{(r)} \left(Z_i - \eta^{(r+1)}\right)^2.$$

The second expression is a responsibility-weighted sample variance with a finite-sample correction. The updated log-scale parameters are then converted to the natural-scale SI mean m and standard deviation s using the transformations defined in Section 3.1. The construction of the zero-safe value T_i^* is explained separately in Section 3.3.3.

The rationale for restricting this parameter update to the PS pathway is epidemiological. A PS observation represents one direct transmission generation, so the PS component follows the target lognormal SI distribution directly. In contrast, the CP component is based on the retained half-normal construction, while the PT and PQ components represent longer transmission chains and rely on working approximations. A parameter update based on all pathways would require the shared SI parameters to improve the fit of the CP, PT, and PQ components simultaneously. Approximation error or poor tail fit in these components could then pull the estimated SI parameters away from the distribution of direct transmission intervals.

The PS-only update instead concentrates parameter learning on the component that contains the clearest one-generation information. An observation contributes to the parameter update in proportion to its PS responsibility, $a_i^{(r)}$. Long ICC intervals that are more compatible with

PT or PQ transmission will generally receive smaller PS responsibilities and will therefore have less influence on the estimated SI mean and standard deviation. Similarly, very short intervals that are more compatible with co-primary transmission will be downweighted in the PS update.

This does not require the SI parameters to reproduce every mode of the complete ICC distribution equally well. In particular, the estimation of m and s is not driven directly by the need to fit the higher-generation PT and PQ peaks. Instead, these components help classify observations in the E-step, while the PS-weighted observations provide the primary information for updating the target SI distribution.

The other pathways are not completely excluded from the estimation process. Their likelihoods and mixture weights remain part of the E-step and therefore affect the PS responsibilities indirectly. If a PT or PQ observation is incorrectly assigned substantial PS responsibility, it can still influence the SI parameter update. Nevertheless, the PS-only construction substantially reduces the direct influence of long-path observations and their tail-shape approximations on the estimation of m and s .

Because the update uses responsibility-weighted moments rather than directly maximizing the complete interval-censored four-pathway likelihood, its purpose is to prioritize direct-transmission information while avoiding the computational burden and potential approximation sensitivity of a full-path numerical parameter update.

7.3.3 Resolving the logarithmic singularity at $T_i = 0$

Zero-day ICC intervals have genuine epidemiological meaning, but the lognormal M-step encounters the singularity

$$\log(0) = -\infty.$$

An uncontrolled near-zero pseudo-observation produces an extremely negative log value. With non-negligible PS responsibility, it pulls $\hat{\eta}$ downward and inflates $\hat{\zeta}$; the resulting PS density becomes flatter, changes the next set of responsibilities, and can assign excessive PS weight to observations that are not direct transmissions. This creates a pathological positive-feedback loop:

$$T_i = 0 \longrightarrow \text{extreme negative log value} \longrightarrow \hat{\zeta} \text{ inflates} \longrightarrow \text{PS density flattens} \longrightarrow \tau_{\text{PS},i} \text{ is distorted} \longrightarrow \text{further vari}$$

The adopted solution is an E/M-decoupled zero-handling strategy. In the E-step, a zero remains a zero and contributes its exact rounded-interval likelihood,

$$p_g(0; m, s) = \int_0^1 2(1 - c) f_g(c; m, s) dc.$$

Only in the logarithmic M-step is the working value replaced by

$$T_i^* = 1, \quad \log(T_i^*) = 0.$$

The value one is a neutral log-scale proxy: it prevents a mathematical singularity without turning a same-day observation into an arbitrarily extreme negative log value. It is not a claim that the true continuous ICC interval was one day. This separation preserves the observation's zero-day meaning in pathway classification while breaking the feedback mechanism that otherwise inflates the variance.

7.3.4 Joint convergence criterion

Pathway weights and distributional parameters are mutually dependent. A small change in m or s alters the component likelihoods; those likelihoods alter responsibilities and weights; the changed weights then affect the next parameter update. Stability of only one or two parameters can therefore be misleading.

At iteration r , the implementation monitors relative changes in the natural-scale parameters,

$$\Delta_m^{(r)} = \frac{|m^{(r+1)} - m^{(r)}|}{m^{(r)} + \epsilon}, \quad \Delta_s^{(r)} = \frac{|s^{(r+1)} - s^{(r)}|}{s^{(r)} + \epsilon},$$

together with the maximum absolute pathway-weight change,

$$\Delta_w^{(r)} = \max_g |w_g^{(r+1)} - w_g^{(r)}|.$$

Convergence is declared only when

$$\Delta_m^{(r)} < \text{tol}, \quad \Delta_s^{(r)} < \text{tol}, \quad \Delta_w^{(r)} < \text{tol}.$$

The default tolerance is 10^{-6} . Including $\Delta_w^{(r)}$ prevents premature termination when the natural-scale parameters appear locally stable but the latent pathway allocation is still changing.

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